

APEX: Automated Pipeline for Explorative Drug Discovery

A target-agnostic QSAR screening and molecular docking platform

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Computational Drug Discovery

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The Drug Discovery Challenge

Scale of the Problem

- 12–15 years from target to approval
- \$2.6 billion average per approved drug
- ~1 in 10,000 screened compounds reaches trials

Computational Acceleration

Early-stage virtual screening reduces the wet-lab search space from millions to tens of candidates before any synthesis is attempted.

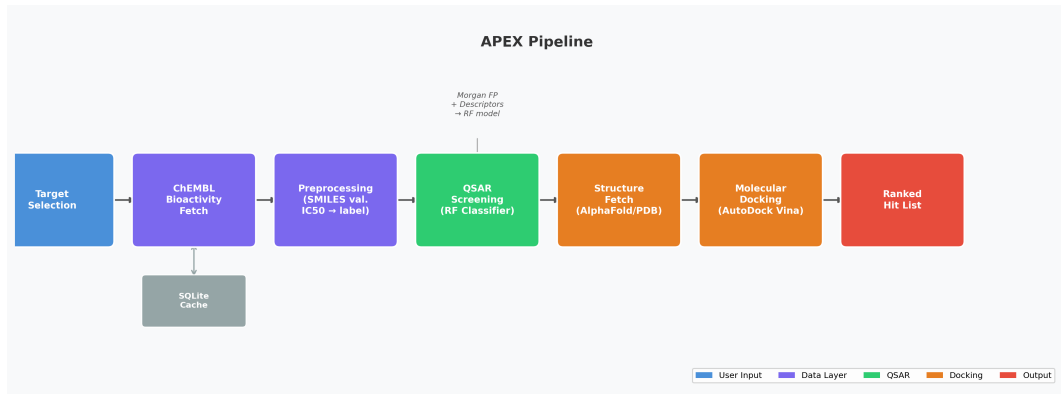
Bottleneck

Manual, target-specific pipelines require expert setup for every new biological target — not scalable.

Our Approach

APEX automates the full pipeline: data ingestion → QSAR → docking for *any* ChEMBL target, on demand.

APEX End-to-End Workflow



Each stage is independently cached and testable; the web platform exposes every stage through a REST API.

ChEMBL Bioactivity Layer

ChEMBL Database

- >19 million bioactivity records
- >2 million unique compounds
- Queried via REST API per target on demand

SQLite Cache

All API responses cached locally (7-day TTL). Re-runs for the same target return in milliseconds.

Data Cleaning Pipeline

- 1 Fetch IC50 (human assays, nM units)
- 2 Validate SMILES via RDKit MolFromSmiles
- 3 Deduplicate by Morgan fingerprint (catches tautomers & stereoisomers)
- 4 Drop grey zone: 1,000–10,000 nM (ambiguous labels)

Label Noise Reduction

Compounds with IC50 in the 1–10 μ M range are removed before training: they are neither clearly active nor clearly inactive and inflate decision-boundary ambiguity in the binary classifier.

QSAR Model Architecture

Molecular Featurisation

Morgan Fingerprints (radius = 2, 1024 bits)
Circular substructure encoding, invariant to SMILES notation.

Physicochemical Descriptors (6 features):

- Mol. weight, LogP
- H-bond donors (HBD), H-bond acceptors (HBA)
- Rotatable bonds, aromatic ring count

Feature vector: $\mathbf{x} \in \mathbb{R}^{1030}$

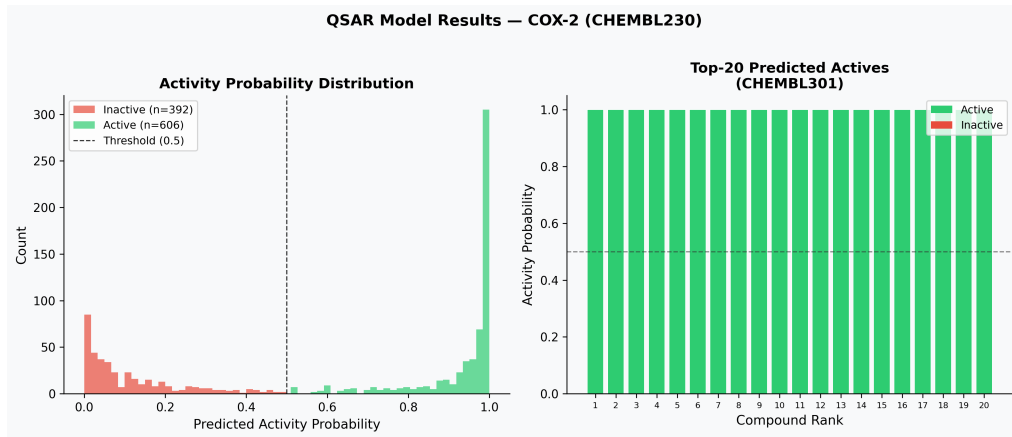
Random Forest Classifier

- 200 trees, random_state=42
- 5-fold stratified cross-validation
- Metrics: CV ROC-AUC and held-out F1
- Serialised to {target_id}.pkl after training

Why Classification?

Cross-assay IC50 noise degrades regression more than classification. `predict_proba` already provides a continuous ranking signal.

QSAR Screening Results



Left: predicted activity probability distributions by true label. *Right:* top-20 ranked compounds coloured by ground-truth activity. CDK2 (ChEMBL301) — CV AUC = 0.943, Test F1 = 0.901, $n_{\text{train}} = 798$.

Candidate Pre-selection: Drug-likeness & Diversity

Step 1 — Lipinski Rule of Five

Drug-likeness gate on predicted actives:

- $MW \leq 500 \text{ Da}$
- $\text{LogP} \leq 5$
- $\text{HBD} \leq 5, \text{ HBA} \leq 10$
- $\text{TPSA} \leq 140 \text{ \AA}^2$

Falls back to all actives if none pass.

Step 2 — Max-Min Diversity

Greedy farthest-point on Tanimoto distance:

- 1 Seed: highest- p drug-like compound
- 2 Add compound most dissimilar to all selected
- 3 Stop at $N = 20$

$$d_T(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}$$

Why Diversity?

Similar scaffolds share binding modes — diversity maximises binding-pocket exploration and avoids redundant docking runs.

AutoDock Vina: Molecule Preparation

Ligand Preparation

- 1 SMILES $\rightarrow$ 3D conformer (RDKit ETKDGv3)
- 2 Force-field minimisation (MMFF94)
- 3 PDBQT conversion (Open Babel `--gen3d`)

Receptor Preparation

- 1 Structure fetch: AlphaFold $\rightarrow$ RCSB PDB
- 2 PDBQT conversion (Open Babel `-xr`)
- 3 Binding box (BioPython):
 - HETATM ligand centroid if co-crystal present
 - Protein geometric centroid otherwise
 - + 10 Å padding, capped at 30 Å

AutoDock Vina: Scoring & Search

Vina Scoring Function

Combines empirical energy terms:

- Steric (Gaussian) repulsion/attraction
- H-bond energy
- Hydrophobic contacts
- Torsional entropy penalty

Output: ΔG_{bind} (kcal/mol)

More negative = tighter predicted binding.

$\Delta G \leq -9$ kcal/mol $\approx$ low-nM K_d

Search Parameters

Exhaustiveness 8 (default Monte Carlo steps)

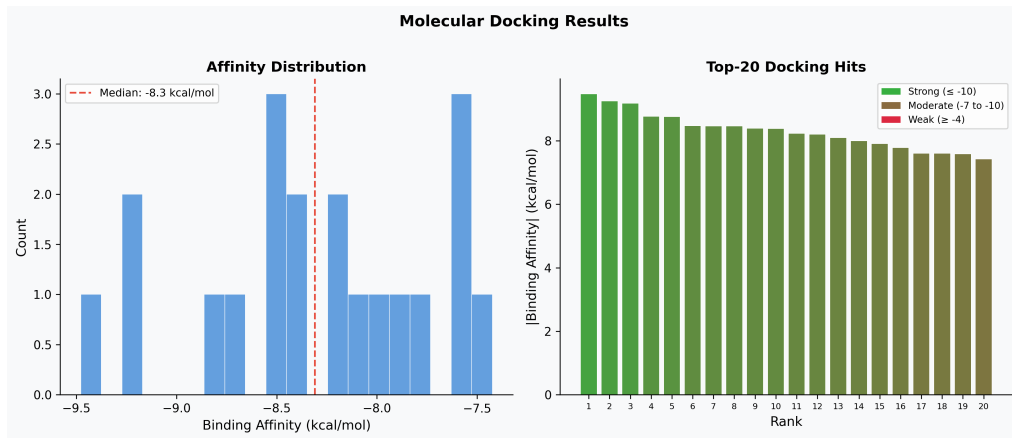
Poses Up to 5 per ligand

Box size 22.5 Å (co-crystal) or adaptive (centroid fallback)

Limitation

Vina scores correlate with experimental K_d in benchmarks but are a scoring approximation, not a thermodynamic calculation. False positives occur, especially with AlphaFold receptors.

Docking Results



Left: binding affinity distribution across top-20 candidates. *Right:* ranked hits coloured by binding strength (green ≤ -9 kcal/mol, yellow moderate, red ≥ -6 kcal/mol). CDK2 — best: -9.5 kcal/mol, median: -8.2 kcal/mol, 3/20 below -9 .

Case Study — Cyclin-Dependent Kinase 2 (CDK2)

Target Profile

ChEMBL ID CHEMBL301

UniProt P24941

Function Cell cycle regulator (G1/S transition)

Disease link Cancer (overexpression in many tumours)

Records 998 IC50 measurements retrieved

QSAR Outcome

- CV ROC-AUC: **0.943** $\pm$ 0.012
- Test F1: **0.901**
- Training time: < 30 s

Docking — Top Hit

Best binding affinity: **−9.5 kcal/mol**

Affinity range of top-20: −9.5 to −7.4 kcal/mol

Compounds with $\Delta G \leq -9$ kcal/mol:
3/20

Total Pipeline Runtime

Stage	Time
ChEMBL fetch	~110 s
QSAR training	<30 s
Docking (20 lig.)	~5 min

Backend — FastAPI

- GET /targets/search?q= — ChEMBL lookup
- POST /qsar/{target_id} — train + predict
- GET /structures/{uniprot_id} — AlphaFold/PDB
- POST /docking/ — Vina run, ranked results

Optional `box_center/box_size` override lets researchers supply known binding-site coordinates.

Frontend — React

- Sidebar: target search & run history
- Real-time pipeline status log
- QSAR table sortable by activity probability
- Docking hit ranking with affinity heat colours
- Compound compare overlay

No Build Step

Single `index.html` with CDN React. Served as a static file by the FastAPI backend on the same origin — no CORS

Summary & Future Work

What APEX Delivers

- Target-agnostic pipeline: ChEMBL → QSAR → docking
- RF classifier on $\mathbb{R}^{1030}$ (Morgan + 6 descriptors)
- Lipinski filter + max-min diversity pre-selection
- AutoDock Vina with adaptive binding-box estimation
- Virtual screening: user-supplied compound libraries
- REST API + interactive web frontend, no build step

Planned Extensions

- Enamine REAL library integration (7 B compounds)
- Ensemble QSAR: XGBoost, graph neural nets
- ADMET filtering pre-docking
- 3D binding pose viewer in-browser
- Automated binding-site detection (fpocket / P2Rank)
- Multi-target comparative screening

Questions?

Source code available on request.

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